

Failure-Driven Substrate Discovery for ARC-AGI-3

A receipt-governed human-in-the-loop development study across public game ontologies

Abstract

We report a linked Kaggle submission with Kaggle public score 0.0, alongside public-scorecard level-1 completions on two structurally distinct ARC-AGI-3 games: ls20 and tr87. This separation is intentional. The submitted package is scoped to two identified public-game substrates and should not be expected to score on unseen ontologies. The paper's contribution is therefore not a general ARC-AGI-3 agent, but a receipt-governed human-in-the-loop development discipline for identifying the substrate a game requires when its state, action, and target ontology are unknown.

This is a human-accelerated development study, not a claim of autonomous substrate discovery. Diagnostic traces exposed substrate gaps; in some cases, source inspection was used inside a disclosed research boundary to validate or instantiate missing mechanics. The contribution is the receipt discipline that prevents such source-informed artifacts from being promoted into unsupported claims of autonomous ontology discovery, general ARC-AGI-3 capability, or private-eval performance.

The development discipline turns failed interactive runs into auditable substrate gaps. It records diagnostic traces, separates observed/refuted/unknown claims, builds the smallest ontology supported by evidence, locally verifies candidate plans, and performs at most one official execution per gated attempt. On ls20, this process produced a substrate-derived level-1 completion with public-scorecard score 3.57, 1 level completed, and 13 actions (App. A, C22). On tr87, after building a separate selector/sequence ontology and closing action-effect coverage with a targeted probe, it produced a level-1 completion with public-scorecard score 4.76, 1 level completed, 14 actions, zero prediction mismatches, and complete provenance (App. A, C36, C37).

The central finding is that the development discipline transferred while the game ontology did not. It produced official level-1 capability on ls20 and tr87, then characterized a third ontology in cn04: click-selected sprite manipulation with movement, rotation/cycling, masking, and marker-completion (App. A, C27, C38, C39). The linked Kaggle score 0.0 is consistent with that thesis: a deliberately ontology-specific package is not evidence of private-eval generality. The public scorecards show what the development discipline produced when the substrate was identified and instantiated; the Kaggle ledger records the current competition score honestly (App. A, C01, C17, C40).

Claim Boundary

This paper claims that a receipt-backed, human-in-the-loop failure-analysis loop can convert ARC-AGI-3 diagnostic failures into bounded substrate knowledge and scoped public-game capability. It demonstrates this with level-1 public scorecard completions on ls20 and tr87, and an audit-only ontology characterization on cn04. **Explicit non-claims:** this work does not claim a general ARC-AGI-3 agent, autonomous substrate discovery, positive Kaggle leaderboard capability, private-eval capability, or all-level performance on any game.

Introduction

ARC-AGI-3 agents can fail before planning begins. A system can explore broadly and still use the wrong kind of world model: a motion heuristic on a selector game, an actor extractor on a click-selected sprite composition task, or a target-matching goal detector where completion is sequence validation. This paper studies that failure mode: substrate-identification failure.

ARC-AGI-3 is interactive. Agents must explore novel environments, infer goals, build models, and execute action-efficient plans without explicit natural-language instructions [ARCAGI3Track; ARCAGI3Benchmark2026]. Official scoring is based on completion and action efficiency, not on internal reasoning volume [ARCAGI3Methodology]. Planning is therefore downstream of a more basic question: what state, actions, and goals are available to be planned over?

A substrate is the set of game-specific facts a planner needs before planning has a coherent target: what counts as actor state, which action effects are stable, what a goal looks like, which frame conventions can be trusted, and when an observed mismatch is a repairable format issue rather than a game-mechanic gap. The method begins with diagnostic runs and treats every mismatch as a claim-bound event. Evidence is triaged into observed, refuted, and unknown claims. A capability run is allowed only after perception, action effects, target semantics, and local verification are specific enough to justify one official attempt.

This is not a high-score leaderboard paper or an autonomous-discovery paper. The Paper Prize requires a linked Kaggle code submission and uses that linked score for the Accuracy category [ARCPrize2026Paper]. Our linked Kaggle submission succeeded with Kaggle public score 0.0 (App. A, C01). That score is not hidden or reinterpreted; it marks the current generalization boundary of a deliberately scoped substrate package. The paper separately contributes an auditable human-in-the-loop development method, two scoped public scorecard completions, and one additional third-game ontology audit: ls20 level 1, tr87 level 1, and cn04 substrate characterization (App. A, C22, C36, C39). Accuracy, in the Kaggle ledger, is weak. Theory, Progress, Completeness, and scoped cross-ontology applicability are where the evidence is strongest.

Related Work

The ARC-AGI-3 benchmark paper defines the setting: interactive environments where agents must explore, infer goals, build internal dynamics models, and plan without natural-language instructions [ARCAGI3Benchmark2026]. This paper does not introduce those benchmark requirements. It studies a receipt-backed development discipline for identifying the substrate a planner is allowed to use.

Recent work on executable world models occupies a different and broader capability lane. "Executable World Models for ARC-AGI-3 in the Era of Coding Agents" reports a coding-agent system that builds executable Python world models, verifies them against observations, refactors toward simpler abstractions, and plans through the model. It reports substantially broader public-game performance than this paper [Rodionov2026EWM]. This paper is not a smaller version of that result. Executable-world-model work demonstrates raw capability through constructed models; this paper studies the prior question of how to identify what substrate should be modeled when the ontology is unknown. Its empirical contribution is the boundary between what transfers across games, the development discipline, and what does not, the per-game ontology. Whereas executable world models show that mechanics can be encoded and planned through once a representation is available, this paper studies the receipt-governed process for deciding which representation is claim-supported enough to use.

Graph-based exploration and map-then-act work similarly show that systematic state/action coverage and pre-execution mapping are strong baselines [RudakovEtAl2025Graph; LiuEtAl2026MAP]. This paper does not claim those ideas as new. Its narrower contribution is the receipt-backed gate from sparse coverage to targeted probes to planner attempts, and the stop rule that blocks further official reruns when a gap becomes architectural rather than another bounded format repair (App. A, C26).

The novelty boundary is therefore:

- failure-driven substrate discovery backed by receipts,
- an explicit claim inventory with observed/refuted/unknown tiers,
- cross-game evidence that the development discipline transferred while ontology did not,
- a substrate-gap taxonomy that distinguishes bounded frame-format gaps from larger game-mechanic gaps, and
- coverage-gated progression from sparse action effects to targeted probes to official planner attempts.

Method: Failure-Driven Substrate Discovery

The method has six steps.

1. Run a bounded diagnostic or substrate attempt and preserve receipts.
2. Extract candidate state/action/target structure from frames and source where allowed.
3. Classify each claim as observed, refuted, or unknown.
4. Build the smallest ontology that the evidence supports.
5. Locally verify a planned action sequence before official execution.
6. Run officially once, then stop on completion, mismatch, provenance break, or

a named substrate gap.

Algorithm 1: Failure-Driven Substrate Discovery

Step	Operation
Input	game instance <i>g</i> , interaction budget <i>B</i> , and claim ledger <i>L</i>
Output	substrate <i>S</i> , capability receipt <i>R</i> , or named gap <i>G</i>
1	Execute a bounded diagnostic probe or substrate attempt under budget <i>B</i>
2	Emit receipt <i>R</i> containing frames, actions, score, state hashes, runner version, and provenance
3	Triage observations into <i>L_obs</i> , <i>L_refuted</i> , and <i>L_unknown</i>
4	Construct the smallest substrate <i>S</i> satisfying <i>L_obs</i> and not contradicting <i>L_refuted</i>
5	If action-effect coverage is below the pre-planner gate, execute a targeted structural probe or stop with <i>coverage_gap</i>
6	If local verification fails, stop with <i>verifier_gap</i> or <i>substrate_gap</i>
7	Authorize at most one official environment execution after the gate opens
8	On success, return a capability receipt; otherwise classify the failure as <i>frame_format_gap</i> , <i>perception_ontology_gap</i> , <i>game_mechanic_gap</i> , or <i>architecture_review_gap</i>

The method intentionally separates score evidence from methodology evidence. Score evidence comes from official scorecards, Kaggle submissions, and environment actions. Local analysis, source inspection, and test execution support the paper's Theory and Completeness claims, but they do not count as ARC-AGI-3 actions and do not themselves create Kaggle-linked Accuracy evidence [ARCPrize2026Paper; ARCAGI3Methodology].

Source-Informed Evidence Boundary

Source-informed work is allowed only inside a disclosed research boundary. In this project, source inspection informed substrates such as ls20 wall geometry, tr87 validation semantics, and the source-static cn04 audit; it also served as an oracle for local verification and source-consistency checks (App. A, C23, C31, C39). It did not authorize private-eval generality, autonomous-discovery claims, or source-derived action replay as a planner claim. That distinction matters because *run_003* and *run_005* produced the same ls20 level-1 action sequence, but they carry different derivation claims: *run_003* was source-derived, while *run_005* recorded object-substrate planner provenance (App. A, C23).

Derivation and status ledger:

Result / Artifact	Interaction-derived?	Source-informed?	Official status
Kaggle Record	N/A	submitted package includes source-informed components	official Kaggle submission
ls20 run_005	partly	yes, geometry	official capability run
tr87 run_001	partly	yes, validation	official capability run
gamma2c .1 selector probe	yes	source-checked	non-capability coverage probe
cn04 gamma3a	limited	yes	no execution run
ls20 level-2 review	limited	yes	no execution run

Allowed-claim ledger:

Result / Artifact	Kaggle linkage	Allowed paper claim
Kaggle Record	Kaggle public score 0.0	linked Kaggle baseline
ls20 run_005	N/A	level-1 public-game completion
tr87 run_001	N/A	level-1 public-game completion
gamma2c .1 selector probe	N/A	coverage gate verification, not score capability
cn04 gamma3a	N/A	ontology characterization only
ls20 level-2 review	N/A	architecture-review gap, no rerun authorized

Automation boundary. In the current implementation, diagnostic execution, receipt capture, selected consistency checks, local verification, and official planner execution are scripted. Substrate interpretation, source-informed mechanic identification, and cross-ontology model repairs are human-authored development steps. The paper therefore claims a receipt-governed

human-in-the-loop development discipline, not an autonomous online agent that independently induces each substrate from environment interaction alone.

Threat model. The main risks are source-informed artifacts being mistaken for autonomous discovery, local verification being mistaken for official score evidence, repeated official attempts becoming informal leaderboard search, and third-game audits being mistaken for capability claims. The mitigation is the claim ledger: each artifact is assigned an evidence type, a derivation boundary, an official/nonofficial status, and an allowed claim.

System Architecture / Receipt Discipline

Every major result is backed by artifacts and tests. The claim inventory is the main safety rail for object-substrate and cross-game evidence added after the early diagnostic runs. Each claim lists artifacts, tests, and non-claim siblings.

The paper lane keeps diagnostic evidence, planner-lane evidence, object-substrate evidence, public scorecard evidence, and Kaggle linkage evidence separate. This prevents a local or source-informed step from becoming an unearned score claim. It also makes productive failures usable: each failed attempt either tightens a substrate, refutes a hypothesis, or names a gap that blocks another run.

The tr87 sequence is the cleanest example of coverage gating. Gamma2c found only 4/20 action/state pairs and blocked planner work (App. A, C32). Gamma2c.1 used a single targeted selector probe to reach 20/20, then gamma2d planned and completed level 1 (App. A, C33, C36). This is not trial-and-error until success; it is process evidence that the coverage gate changed the decision boundary before the official planner attempt. It is not claimed as a controlled ablation.

ls20 Case Study

The ls20 substrate began as a motion/object ontology. The actor state tracked position, shape, color, and rotation. Early diagnostic traces found motion correlations and frame-shape anomalies but did not justify a goal detector (App. A, C02-C09).

The first successful object-substrate capability result was `run_005`. It completed level 1 with public-scorecard score 3.57, one level completed, and 13 actions after local verification (App. A, C22). The sequence matched the earlier source-derived handoff sequence, but the claim boundary is different: `run_003` was source-derived while `run_005` was planned by the object substrate and preserved planner-derived sequence provenance (App. A, C23).

Later ls20 work exposed the method's stop condition. `run_008` crossed the level-1 transition online but still completed only one level and recorded `official_local_online_divergence` (App. A, C24). The subsequent source mechanic review emitted `architecture_review_required` because the dynamic trigger-bound sprite automata failed the bounded-primitive gate (App. A, C25).

That finding is part of the result, not a footnote. The method did not run `run_009` from momentum. It classified the gap as larger than another frame-format patch and pivoted to cross-game transfer.

tr87 Case Study

Gamma2a tested whether the ls20 substrate transferred to tr87. It did not. The transfer audit classified the result as `method_transfers_ontology_doesnt`: frame-shape handling transferred, while actor extraction, action effects, target extraction, and game mechanics required tr87-specific redesign (App. A, C27).

Gamma2b then built a tr87-specific perception layer. It extracted well-formed `Tr87ActorState` on 81/81 probe frames (App. A, C28). The target structure was sequence-rule matching, not ls20-style position plus shape/color/rotation equality (App. A, C29).

Gamma2c built a separate action-effect model. It analyzed 80 transitions and identified selector-position and sequence-state effects, explicitly not motion deltas (App. A, C30). The consistency checker passed 80/80 source checks with mismatch count zero (App. A, C31).

The planner did not run immediately. Gamma2c recorded only 4/20 action/state pairs and `ready_for_gamma2d_planner : false` (App. A, C32). A targeted selector probe then completed the coverage objective with 20/20 action-state pairs and source consistency 1.0, while scoring 0.0 because it was not a capability attempt (App. A, C33, C34).

After that gate opened, gamma2d planned over selector/sequence state, locally verified a 14-action plan, and completed tr87 level 1 officially with public-scorecard score 4.76, one level completed, and 14 actions (App. A, C35, C36). The receipt recorded zero prediction mismatches and complete provenance (App. A, C37).

Cross-Game Results

The cross-game result is qualitative and scoped. The same method produced official level-1 completions in two games, and then characterized a third game ontology without attempting capability. The capability ledger is still two games. The method-transfer ledger is now three ontologies.

Game	Ontology Characterized	Substrate Status	Capability Status	Claim Boundary
ls20	motion plus shape/color/rotation target matching	object substrate with wall geometry and bundle handling	official level-1 completion, public-scorecard score 3.57	not level 2+, not private-eval generality
tr87	selector index plus sequence-rule construction	tr87-specific perception, action effects, coverage probe, local verifier	official level-1 completion, public-scorecard score 4.76	not all-level tr87 capability
cn04	click-selected sprite manipulation with movement, rotation/cycling, masking, and marker-completion	source-static substrate audit and transferability map	no official run, no planner, no capability claim	method-characterization evidence only

Scores are rounded in the main body text; exact machine values are preserved in the claim inventory and receipts.

ls20 used actor position, shape/color/rotation, wall geometry, frame-bundle handling, and exact target matching. tr87 used selector index, sequence slots, tile variant cycling, source-consistent action effects, and sequence-rule target validation (App. A, C38). Gamma3a selected cn04 because its source exposes a third ontology: click selection (ACTION6), selected sprite movement (ACTION1 through ACTION4), rotation or overlapping-sprite cycling (ACTION5), masking, and marker-completion (App. A, C39).

This is not a third-game completion or all-level generality claim. It is a cross-ontology development-discipline claim: three structurally different public games were characterized by the same evidence-gated pipeline, and two of those substrates produced official level-1 completions. The shared contribution is the discipline: diagnose, model, gate on coverage, verify locally when a planner is attempted, run once, and stop on named gaps. The shared contribution is not a single game ontology.

The earlier N=2 weakness therefore changes shape. Capability remains N=2, but method characterization is N=3 (App. A, C39). This paper treats this as stronger evidence for cross-ontology applicability of the development discipline, not as proof of broad ARC-AGI-3 generality.

Substrate-Gap Taxonomy

The artifact chain supports two operational gap classes (App. A, C26).

Frame-format gaps occur when the runner emits a frame or bundle convention the extractor handles incorrectly while the underlying mechanic remains within the current planner ontology. In the ls20 chain, multi-frame transition bundles and grayscale bundle shapes were bounded gaps: each could be named, repaired, tested, and re-run without changing the planner's game-mechanic model.

Game-mechanic gaps occur when the current state/action model lacks a mechanic needed for planning. ls20 level 2 is the current example. The source review found dynamic trigger-bound sprite automata and budgeted event semantics that failed the bounded-primitive gate (App. A, C25).

Schematically, the ls20 level-2 review emitted a stop decision with these fields:

- decision: architecture_review_required
- gap_class: game_mechanic_gap
- reason: dynamic trigger-bound sprite automata exceeded the

bounded-primitive gate

- official_rerun_authorized: false

The taxonomy is not claimed as an exhaustive law of ARC-AGI-3. Its value is operational: it predicts what kind of next action is justified. Frame-format gaps can often justify a small substrate repair and one verified rerun. Game-mechanic gaps may require architecture review, local event-state modeling, or a pivot. In this chain, the taxonomy's practical test was that it blocked another ls20 run when the failure was no longer a bundle-selection problem.

Limitations

The linked Kaggle submission succeeded and records Kaggle public score 0.0 (App. A, C01). The two positive scores are public scorecard evidence on covered public games, not Kaggle leaderboard or private-eval claims (App. A, C17, C40). This paper does not ask the reader to treat public scorecards as a substitute for the Kaggle ledger; it asks the reader to evaluate a methodology whose current Kaggle Accuracy evidence is weak and honestly recorded.

The capability evidence is level-1 only in two games. ls20 level 2 remains an architecture-review gap (App. A, C25), and tr87 levels beyond level 1 are untested. The cross-ontology evidence is stronger after gamma3a, but it is still asymmetric: method characterization spans ls20, tr87, and cn04, while official completion evidence remains ls20 and tr87 only (App. A, C39).

Several components are source-informed. They are valid local research evidence, and the submitted package records its licensing and provenance posture, but they do not support private-eval generality or a broad competition framing. They also do not support a claim that the agent autonomously discovered each substrate. The paper should be judged on the scoped claim it makes: receipt-backed human-in-the-loop substrate discovery produced official level-1 public scorecard completions in two different ontologies and named the point where further ls20 iteration became architectural.

Conclusion

This project began with diagnostic traces that scored 0.0. It now has a guarded evidence chain: diagnostic receipts, claim tiers, object-substrate modules, coverage gates, local verifiers, a linked Kaggle submission with Kaggle public score 0.0, scoped public scorecard completions in two games, and a third ontology audit for cn04.

What transferred was the development discipline: diagnostic traces, claim tiers, coverage gates, local verification before official execution, and stop rules when a gap became architectural. What did not transfer was the game ontology. ls20 needed motion, object attributes, wall geometry, and target matching. tr87 needed selector index, sequence state, and rule validation. cn04 exposed click-selected sprite manipulation, rotation/cycling, masking, and marker-completion.

The main result is not that one ontology is sufficient for ARC-AGI-3. The main result is that planning is downstream of substrate identification. Failure traces can reveal which substrate is missing; local verification can decide when a capability attempt is justified; and official runs should stop when the next gap is no longer bounded. In the current evidence, that discipline produced ls20 and tr87 level-1 completions and characterized cn04 as a third distinct ontology while showing that the development discipline transferred across the studied public ontology types and the game ontology did not.

The method does not remove the need for game-specific modeling. It makes that need explicit, testable, and governed by stop rules.

Data and Reproducibility Availability

The source repository for the submitted method is <https://github.com/Haserjian/arc3-substrate-discovery-paper>. The linked Kaggle notebook is <https://www.kaggle.com/code/timhaserjian/arc3-object-substrate-submission-candidate>. Internal evidence is cited by repo-relative artifact paths rather than local filesystem paths; the submission candidate contains no local home-directory absolute paths.

Key repo-relative evidence paths:

- artifacts/kaggle_submission_preflight/kaggle_submission_record/submission_record.json
- artifacts/object_substrate_v0/run_005_object_aware_wall_fix/receipt.json
- artifacts/object_substrate_v0/tr87_run_001_object_aware/receipt.json
- artifacts/object_substrate_v0/gamma3a_third_game_substrate_audit/game_selection.json
- artifacts/object_substrate_v0/gamma3a_third_game_substrate_audit/gamma3b_decision.json
- artifacts/object_substrate_v0/gamma3a_third_game_substrate_audit/transferability_map_vs_ls20_tr87.json
- docs/object_substrate_v0/gamma3a_third_game_substrate_audit.md
- docs/paper/cross_game_substrate_results.md
- docs/paper/substrate_gap_taxonomy.md
- docs/paper/claim_inventory.md
- docs/paper/results_table.md
- docs/paper/known_gaps.md

- artifacts/object_substrate_v0/gamma2a_cross_game_substrate_report/cross_game_substrate_report.json
- artifacts/object_substrate_v0/run_008_grayscale_bundle_fix/level2_source_mechanic_review/next_decision.json
- artifacts/object_substrate_v0/run_008_grayscale_bundle_fix/receipt.json
- artifacts/object_substrate_v0/tr87_action_effect_model/action_effect_model.json
- artifacts/object_substrate_v0/tr87_action_effect_model/consistency_check.json
- artifacts/object_substrate_v0/tr87_action_effect_model/known_gaps.json

Appendix A: Claim Inventory Summary

This appendix is a reader-facing index to the internal claim inventory. It is not a separate score claim; the full inventory in the repository carries artifacts, tests, and non-claim siblings. Reader summaries may shorten long machine identifiers for PDF legibility; exact identifiers are preserved in the repository inventory and receipts.

Claim ID	Tier	Reader summary
C01	OBSERVED	The ARC-AGI-3 Kaggle linkage submission succeeded and recorded Kaggle public score 0.0.
C02	OBSERVED	The ls20 official-runner diagnostic trace scored 0.0, completed 0 levels, and used 81 actions.
C03	OBSERVED	The ls20 rotating probe produced 81 novel states and 0 completed levels.
C04	OBSERVED	ls20 produced cardinal largest-region motion correlations for ACTION1/ACTION2/ACTION3/ACTION4 under medium-confidence action attribution.
C05	OBSERVED	The tested fixed action prefix from reset matched 8/8 paired replay rows.
C06	REFUTED	The hypothesis that tested direction pairs return to the exact before-frame state hash is refuted by 0/4 returns.
C07	REFUTED	The clean position-level inverse hypothesis is refuted for the tested pair with position-level non-return evidence.
C08	OBSERVED	Rows 42/43 in ls20 show frame-shape changes [1, 64, 64] -> [6, 64, 64] and [6, 64, 64] -> [1, 64, 64].
C09	UNKNOWN	The best current hypothesis is that rows 42/43 may reflect a structural/render-phase event, but the semantic label is unknown.
C17	REFUTED	Current artifacts explicitly refute positive Kaggle leaderboard capability and private-eval capability claims; the Kaggle public score is 0.0.
C22	OBSERVED	run_005 completed ls20 level 1 with score 3.571428571428571, 1 level completed, and 13 actions after local verification passed.
C23	OBSERVED	run_005 planned sequence [3, 3, 3, 1, 1, 1, 1, 4, 4, 4, 1, 1, 1] matched the earlier source-derived run_003 sequence, while run_005 recorded planner-derived sequence provenance.
C24	OBSERVED	run_008 locally verified levels [1, 2], crossed the level-1 transition online, but the official run still completed only 1 level and recorded an official/local online-divergence marker.
C25	OBSERVED	The level-2 source mechanic review emitted architecture_review_required because the source-visible dynamic trigger-bound sprite automata failed the bounded-primitive gate.
C26	OBSERVED	The artifact chain supports a paper taxonomy separating bounded frame-format gaps from larger game-mechanic gaps that may require architecture review.
C27	OBSERVED	The gamma2a cross-game substrate audit classified the finding as a method-transfers/ontology-does-not-transfer finding: frame/bundle and diagnostic layers transferred, while actor, action, target, and mechanic layers required tr87-specific redesign.
C28	OBSERVED	The tr87-specific perception audit produced well-formed Tr87ActorState on 81/81 frames with coverage rate 1.0.
C29	OBSERVED	The tr87 target extractor characterized the goal as sequence-rule matching with completion predicate the source-specific sequence-rule validation predicate.
C30	OBSERVED	The tr87 action-effect model analyzed 80 transitions and identified selector-position and sequence-state effects, with no motion-delta representation.
C31	OBSERVED	The gamma2c consistency checker reported 80/80 transition checks passed with pass rate 1.0 and mismatch count 0.
C32	OBSERVED	Before the selector probe, gamma2c recorded only 4/20 observed action/state pairs, 16 known gaps, and planner readiness false.
C33	OBSERVED	The targeted selector coverage probe observed 20/20 (action, selector_index) pairs, with source consistency pass rate 1.0 and planner readiness true.
C34	OBSERVED	The selector coverage probe scored 0.0, completed 0 levels, and took 20 actions while completing its coverage objective.
C35	OBSERVED	The tr87 local verifier passed a 14-action plan over selector/sequence state before official execution.
C36	OBSERVED	The tr87 object-aware planner official run scored 4.761904761904762, completed 1 level, took 14 actions, and used planned sequence [2, 2, 3, 2, 2, 3, 2, 3, 1, 1, 1, 3, 2, 2].
C37	OBSERVED	The tr87 run_001 receipt reported zero prediction mismatches, no failure modes, and complete provenance.
C38	OBSERVED	The object-substrate pipeline produced official level-1 completions for both ls20 and tr87, despite ls20 using position/shape/color/rotation matching and tr87 using selector/sequence-rule matching.
C39	OBSERVED	Gamma3a characterized cn04 as a third distinct ontology, click-selected sprite manipulation with movement, rotation/cycling, masking, and marker-completion, while explicitly emitting audit_only_sufficient rather than authorizing a planner or official run.
C40	REFUTED	The current evidence refutes broad claims that the project has a general ARC-AGI-3 agent, private-eval capability, or positive Kaggle leaderboard performance.

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